



PetPaws

Pet Foods & Accessories Store

MERN Stack — MongoDB · Express.js · React.js · Node.js

**MERN Stack E-Commerce
Application**

Project Name: PetPaws — Pet Foods & Accessories Store

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INTRODUCTION

PetPaws is your one-stop destination for effortless online pet shopping. With a user-friendly interface and a comprehensive product catalog covering pet food, treats, toys, grooming products, cages, leashes, collars, beds, health supplements, and all pet-related accessories, finding the perfect items for your beloved companions has never been easier.

Seamlessly navigate through detailed product descriptions, customer reviews, and available discounts to make informed purchasing decisions. Enjoy a secure checkout process and receive instant order confirmation. For administrators, our robust dashboard provides efficient order management, insightful analytics, inventory tracking, and complete product management to drive business growth.

Experience the future of online pet shopping with PetPaws today — because your pet deserves the very best!

SCENARIO

Ravi's New Puppy

Ravi, a first-time dog owner, has just brought home a 3-month-old Labrador puppy named Bruno. Excited but overwhelmed, Ravi needs to quickly source high-quality puppy food, a collar, a leash, a bed, and some chew toys. With a hectic work schedule, Ravi turns to PetPaws to simplify his search and get everything delivered to his doorstep.

1. **Effortless Product Discovery:** Ravi opens PetPaws and navigates to the Dog Food category. He is greeted with a diverse range of puppy nutrition options from trusted brands like Royal Canin and Pedigree. Using the filtering options, Ravi selects 'Puppy Food' under Dog Food and refines his search based on Bruno's age and weight.
2. **Personalized Recommendations:** As Ravi scrolls through the curated selection, he notices a 'Featured Products' section. Intrigued, he clicks on it and discovers a complete starter kit. Impressed by the product ratings and verified purchase reviews, Ravi adds the Royal Canin Puppy Starter to his cart.
3. **Seamless Checkout Process:** With the items in his cart, Ravi proceeds to checkout. He enters his delivery address, selects UPI as his preferred payment method, and completes the transaction in just a few clicks thanks to PetPaws's secure and efficient checkout process.
4. **Order Confirmation:** Moments after placing the order, Ravi receives a confirmation notification from PetPaws with a unique order ID. He can track the real-time status of his order from the My Orders section on his profile.
5. **Efficient Order Management for Admin:** On the admin side, the PetPaws administrator receives a notification of Ravi's purchase through the admin dashboard. They quickly update the order status to 'Confirmed' then 'Shipped', with tracking details added automatically.
6. **Happy Pet Parent:** Within two days, Bruno's food, collar, leash, and bed arrive at Ravi's home. Bruno is thrilled with his new toys, and Ravi is delighted by the seamless experience PetPaws provided from discovery to delivery.

In this scenario, PetPaws proves to be the perfect solution for Ravi's busy lifestyle. From effortless product discovery to secure checkout and efficient order management, PetPaws simplifies the entire process — allowing new pet parents to care for their companions with confidence and ease.

SYSTEM REQUIREMENTS

1. Software Requirements

These are the essential tools and platforms required to develop, test, and run the PetPaws application efficiently.

- **Operating System:** Windows 10/11, macOS, or Linux — Supports cross-platform development and testing.
- **Node.js (v18 or above):** Provides the runtime environment for building and managing the backend API and frontend tooling.
- **npm (v8 or above):** A package manager required to install dependencies for React, Express, and all related libraries.
- **React.js (v18):** A JavaScript library for building dynamic and responsive user interfaces with component-based architecture.
- **Vite (v5):** A fast frontend build tool and development server used to scaffold and serve the React application.
- **Express.js (v4):** A lightweight web framework for building RESTful APIs in Node.js.
- **MongoDB:** A NoSQL database used to store data for users, products, orders, carts, reviews, and admin settings.
- **Mongoose (v8):** An ODM library that provides schema-based modelling for MongoDB collections.
- **Axios:** An HTTP client for making API requests from the React frontend to the Express backend.
- **bcryptjs:** For hashing and securely comparing user passwords in the authentication system.
- **jsonwebtoken (JWT):** For generating and verifying JSON Web Tokens used in authentication and authorization.
- **Browser:** Google Chrome / Firefox (latest version) — For rendering and testing the UI in real time.
- **Postman:** Tool for testing and documenting REST API endpoints during development.
- **Visual Studio Code:** Preferred code editor with built-in Git support, terminal, and extension ecosystem.
- **Git & GitHub:** For version control and collaborative development.

2. Hardware Requirements

Describes the minimum and recommended specifications needed to support the development and usage of the application.

- **Processor:** Intel Core i5 (8th Gen or above) / AMD Ryzen 5 or better — Ensures fast compilation and multitasking during development.
- **RAM:** Minimum 8 GB (16 GB recommended) — For handling development servers, IDEs, and browser testing simultaneously.
- **Storage:** At least 2 GB free space — Required for package installations, MongoDB setup, and local project files.
- **Display:** 1366x768 or higher — Recommended for optimal coding experience and application layout visualization.

PROJECT ARCHITECTURE

Technical Architecture

The PetPaws application follows a classic three-tier MERN stack architecture with a clear separation between the Presentation Layer (React.js), the Business Logic Layer (Node.js + Express.js), and the Data Layer (MongoDB).

In this architecture diagram:

- The Frontend is represented by the React.js section, including UI components such as Home, Products, Product Detail, Cart, Checkout, Wishlist, Profile, Orders, and the Admin Dashboard.
- The Backend is represented by the Express.js + Node.js section, consisting of API endpoints for Auth, Products, Cart, Orders, Users, Reviews, Wishlist, and Admin.
- The Database section represents MongoDB, which stores collections for Users, Products, Orders, Cart, Reviews, and AdminSettings.

FRONTEND (React.js)	BACKEND (Express.js)	DATABASE (MongoDB)
Home Page	Auth Routes & Controller	Users Collection
Products Page	Product Routes & Controller	Products Collection
Product Detail	Cart Routes & Controller	Orders Collection
Shopping Cart	Order Routes & Controller	Cart Collection
Checkout	User Routes & Controller	Reviews Collection
Wishlist	Review Routes & Controller	AdminSettings Collection
User Profile	Wishlist Controller	
My Orders	Admin Controller	
Admin Dashboard	JWT Middleware	
Admin Products	Error Handler	

ER Diagram Overview

The PetPaws ER diagram represents the entities and relationships involved in the pet e-commerce system. It illustrates how users, products, cart, orders, and reviews are interconnected.

- **USER:** Represents the individuals registered on the platform. Contains fields for username, email, password, phone, avatar, role (user/admin), addresses array, wishlist array, and active status.

- **Admin:** Represents the AdminSettings collection with important details such as banner images, product categories, shipping fees, and tax configuration.
- **Products:** Represents the collection of all pet products available on the platform — food, toys, accessories, grooming items, health supplements, and more.
- **Cart:** Stores all products added to cart by users. Each cart is differentiated by userId and contains an array of items with product reference, quantity, price, and variant.
- **Orders:** Stores all orders made by users, including order items, shipping address, payment details, order status history, and tracking information.
- **Reviews:** Stores user-submitted product reviews including star rating, title, comment, and verified purchase status.

FEATURES

PetPaws delivers a comprehensive and feature-rich e-commerce experience tailored specifically for pet owners and pet product administrators.

7. **Comprehensive Pet Product Catalog:** PetPaws boasts an extensive catalog covering Dog Food, Cat Food, Bird Food, Fish Food, Toys, Grooming products, Accessories, Health Supplements, Beds, Cages, Leashes, Collars, and Clothing. Each product has detailed descriptions, images, brand info, weight/size variants, and pet-type filtering.
8. **Advanced Product Search & Filters:** Users can search by keyword and apply multiple filters simultaneously — category, pet type (Dog, Cat, Bird, Fish, etc.), price range, brand, minimum rating, and sort options (Newest, Popular, Top Rated, Price Low-High, Price High-Low) — with paginated results.
9. **Add to Cart & Wishlist:** Each product listing features convenient 'Add to Cart' and 'Add to Wishlist' buttons. Users can update quantities, remove items, move from wishlist to cart, and the cart displays real-time totals with shipping calculations.
10. **Multi-Step Secure Checkout:** PetPaws provides a 3-step checkout process: (1) Shipping Address with saved address selection, (2) Payment Method selection (COD, Card, UPI, Net Banking), and (3) Order Review before final placement. Stock is validated before order creation.
11. **Order Confirmation & Tracking:** After placing an order, users receive an instant confirmation with their Order ID. A visual status timeline (Pending → Confirmed → Processing → Shipped → Delivered) is shown on the Order Detail page along with tracking numbers.
12. **Product Reviews & Ratings:** Customers can submit star ratings (1-5), a review title, and detailed comments. Verified Purchase badges are automatically applied for users who have received the product. Average ratings are recalculated dynamically.
13. **User Profile & Address Management:** Users can update their profile (name, phone), change password securely, and manage multiple saved delivery addresses with a default address designation for faster checkout.
14. **Robust Admin Dashboard:** Administrators get a data-rich dashboard showing total revenue, order count, customer count, product count, monthly revenue bar chart, top-selling products, recent orders, and low-stock alerts.
15. **Admin Product Management:** Admins can create, edit, delete (soft delete), and search products with full control over all fields including category, pet type, price, discount, stock, images, variants, tags, and featured status.
16. **Admin Order Management:** Admins can view all orders with status filter tabs, update order status through the full lifecycle, add tracking numbers, and add internal notes for each status change.
17. **Admin Customer Management:** Admins can view all registered customers, search by name or email, and activate/deactivate user accounts as needed.

18. **Store Settings Management:** Admins can configure shipping fees, free shipping thresholds, tax rates, store name, tagline, contact details, homepage banners, and product category definitions.

ROLES AND RESPONSIBILITIES

USER

Represents the individuals or entities who are registered on the PetPaws platform as pet owners or shoppers.

- Can browse, search, and filter all available pet products.
- Can add products to the shopping cart and wishlist.
- Can complete the checkout process with address and payment details.
- Can view order history, track order status, and cancel eligible orders.
- Can submit product reviews and ratings.
- Can manage their personal profile, password, and multiple delivery addresses.

Admin

Represents the platform administrator who manages all store operations through the Admin Dashboard.

- Can log into the dedicated admin panel with elevated access.
- Can create, edit, and remove pet product listings.
- Can manage product categories, banner images, and site settings.
- Can view and update the status of all customer orders.
- Can monitor inventory levels and receive low-stock alerts.
- Can view all registered customers and manage their account status.
- Can access dashboard analytics including revenue charts and top products.

Products

Represents the collection of all pet products available on the platform, including food, accessories, toys, and health products.

Cart

Stores all products added to the cart by individual users. Each cart is differentiated by the user ID and maintains item-level quantity tracking with real-time price computation.

Orders

Stores all orders placed by users on the platform, including full order item details, shipping address, payment method, status history, and delivery tracking.

User Flow

- Users start by registering for an account with username, email, and password.
- After registration, they can log in with their credentials and receive a JWT token.
- Once logged in, they can browse all available pet products with advanced filters.
- Users can add products to the cart and wishlist from the product listing or detail pages.
- They then proceed through the 3-step checkout: Address → Payment → Review & Confirm.
- After ordering, they can track their orders in real time from the My Orders section.
- Users can submit product reviews after receiving delivered orders.

MVC PATTERN

The PetPaws backend application follows the Model-View-Controller (MVC) architectural pattern, a software design approach that separates the application into three interconnected layers. This separation allows for modularity, easier maintenance, and scalability.

Model Layer (Data Layer)

The Model layer is responsible for handling all data-related logic. This includes the definition of data schemas and the operations performed on the database using those schemas. The models are implemented using **Mongoose**, which provides a schema-based solution to model application data for MongoDB.

PetPaws defines the following Mongoose models:

- **User Model:** Handles user registration data, password hashing, address management, wishlist, and role assignment.
- **Product Model:** Stores all pet product data including category, petType, variants, pricing, stock, and ratings.
- **Order Model:** Manages the full order lifecycle from creation to delivery with status history tracking.
- **Cart Model:** Maintains per-user cart state with product references, quantities, and prices.
- **Review Model:** Stores product reviews with ratings, verified purchase flags, and approval status.
- **Admin Model:** Stores site-wide configuration including banners, categories, and business settings.

Controller Layer

The Controller layer acts as an intermediary between the routes and the models. It receives incoming requests, processes the input (including validation or transformation), calls the appropriate methods from the model, and then returns a response to the client.

PetPaws controllers include:

- **authController.js** — Handles user registration, login, admin login, and token validation.
- **productController.js** — Manages product CRUD, featured products, related products, brand listing, and search.
- **cartController.js** — Handles adding items, updating quantities, removing items, and clearing the cart.

- **orderController.js** — Manages order creation, user order history, admin order listing, and status updates.
- **userController.js** — Handles profile updates, password changes, and address management.
- **reviewController.js** — Manages product review submission, listing, and deletion.
- **wishlistController.js** — Handles toggling wishlist items and moving items to cart.
- **adminController.js** — Provides dashboard statistics, settings management, and banner control.

View Layer (Routing Layer)

In the context of a backend REST API, the **View** is implemented as the **routing layer**, where various endpoints are defined. These endpoints determine how the backend responds to different HTTP requests (GET, POST, PUT, DELETE) and are responsible for invoking the appropriate controller functions.

Advantages of Using MVC in This Project

- **Separation of Concerns:** Each layer has a clearly defined responsibility, improving readability and maintainability.
- **Scalability:** New features can be added easily by creating new routes, controllers, and models.
- **Reusability:** Logic in controllers and models can be reused across multiple parts of the application.
- **Testing:** Each layer can be tested independently, especially controllers and models.
- **Collaboration-Friendly:** Multiple developers can work simultaneously on different layers without conflict.

PROJECT SETUP AND CONFIGURATION

Creating the Project Folder

19. Create a new folder named petpaws.
20. Inside that folder, create two sub-folders.
21. Name one **client**.
22. Name the other **server**.
23. Now open the petpaws folder in VS Code.

Client Setup (Installing React App)

- Open the client folder in the terminal of VS Code.

```
npm create vite@latest . -- --template react
```

- Select React framework from the given options:

```
Select a framework:  
|  React
```

- Select JavaScript variant from the given options:

```
Select a variant:  
|  JavaScript
```

- Navigate to the client folder:

```
cd client
```

- To install all the packages, run:

```
npm install
```

- Install additional required libraries:

```
npm install axios react-router-dom react-hot-toast react-icons
```

- To start the React development server:

```
npm run dev
```

Server Setup (npm init)

- Open the server folder in the terminal of VS Code.

```
npm init -y
```

- **Install server dependencies:**

```
npm install express mongoose bcryptjs jsonwebtoken cors dotenv multer
npm install --save-dev nodemon
```

- **Create the following files and folder structure:**

```
server/
├── index.js
├── .env
├── config/
│   └── db.js
├── models/
│   ├── User.js
│   ├── Product.js
│   ├── Order.js
│   ├── Cart.js
│   ├── Review.js
│   └── Admin.js
├── controllers/
│   ├── authController.js
│   ├── productController.js
│   ├── cartController.js
│   ├── orderController.js
│   ├── userController.js
│   ├── reviewController.js
│   ├── wishlistController.js
│   └── adminController.js
├── routes/
│   ├── authRoutes.js
│   ├── productRoutes.js
│   ├── cartRoutes.js
│   ├── orderRoutes.js
│   ├── userRoutes.js
│   ├── reviewRoutes.js
│   ├── wishlistRoutes.js
│   └── adminRoutes.js
├── middleware/
│   └── auth.js
└── data/
    └── seeder.js
```

BACKEND DEVELOPMENT

1. Setup Express Server

- Create index.js as the application entry point.
- Create an Express server on port 8000.
- Configure CORS middleware to allow requests from the React frontend (port 5173).
- Define and mount all API routes.

```
// index.js
const express = require('express');
const cors = require('cors');
const dotenv = require('dotenv');
dotenv.config();
connectDB();
const app = express();
app.use(cors({ origin: 'http://localhost:5173', credentials: true }));
app.use(express.json({ limit: '10mb' }));
app.listen(process.env.PORT || 8000);
```

2. Database Configuration

- Set up MongoDB either locally or using MongoDB Atlas cloud service.
- Create a config/db.js file to manage the Mongoose connection.
- Define the .env file with the MongoDB connection string.

```
// config/db.js
const mongoose = require('mongoose');
const connectDB = async () => {
  const conn = await mongoose.connect(process.env.MONGO_URI);
  console.log('MongoDB Connected: ' + conn.connection.host);
};
module.exports = connectDB;
```

3. Create Express.js Server

- Set up an Express.js server to handle HTTP requests and serve API endpoints.
- Configure middleware including body-parser equivalent (express.json) and cors.
- Mount all route modules under the /api prefix.

4. Define API Routes

- Create separate route files for each API domain: auth, products, cart, orders, users, reviews, wishlist, admin.

- Define necessary routes for product listing, user authentication, order management, and admin operations.
- Implement route handlers using Express.js to handle requests and interact with the database.

Route Module	Endpoints
/api/auth	POST /register, POST /login, POST /admin/login, GET /me
/api/products	GET /, GET /featured, GET /brands, GET /:id, GET /:id/related, POST /, PUT /:id, DELETE /:id
/api/cart	GET /, POST /, PUT /:itemId, DELETE /:itemId, DELETE /
/api/orders	POST /, GET /my, GET /:id, PUT /:id/cancel, GET / (admin), PUT /:id/status
/api/users	PUT /profile, PUT /password, POST /addresses, PUT /addresses/:id, DELETE /addresses/:id
/api/reviews	POST /, GET /product/:productId, DELETE /:id
/api/wishlist	GET /, POST /:productId, DELETE /:productId
/api/admin	GET /dashboard, GET /settings, PUT /settings, POST /banners, DELETE /banners/:id

5. Implement Data Models

- Define Mongoose schemas for User, Product, Order, Cart, Review, and AdminSettings.
- Create corresponding Mongoose models to interact with the MongoDB database.
- Implement CRUD operations for each model.

6. User Authentication

- Create routes and controllers for user registration, login, and admin login.
- Hash passwords using bcryptjs before saving to database.
- Generate JWT tokens on successful login for session management.
- Set up authentication middleware (auth.js) to protect private routes.

```
// middleware/auth.js - protect middleware
const protect = async (req, res, next) => {
  const token = req.headers.authorization?.split(' ')[1];
  const decoded = jwt.verify(token, process.env.JWT_SECRET);
  req.user = await User.findById(decoded.id).select('-password');
  next();
};
```

```
const adminOnly = (req, res, next) => {  
  if (req.user?.role === 'admin') next();  
  else res.status(403).json({ message: 'Admin access required' });  
};
```

7. Handle Products and Orders

- Create routes and controllers to handle pet product listings, including filtering by category, petType, brand, price range, and keyword search with MongoDB text indexing.
- Implement the ordering functionality with stock validation, automatic discount price calculation, shipping fee computation, and tax calculation.
- After order placement, automatically deduct product stock and increment soldCount.

8. Admin Functionality

- Implement routes and controllers for admin-specific operations including product CRUD, order status management, user management, and store settings.
- Add authentication (protect) and authorization (adminOnly) middleware to all admin routes.
- Implement the admin dashboard endpoint that aggregates revenue, order counts, top products, and low-stock alerts using MongoDB aggregation pipelines.

9. Error Handling

- Implement global error handling middleware to catch and handle all errors during API requests.
- Return appropriate HTTP status codes (400, 401, 403, 404, 500) with descriptive error messages.
- All controllers are wrapped in try-catch blocks to handle database and validation errors gracefully.

DATABASE DEVELOPMENT

Schema Use-Case

1. User Schema

- **Schema:** userSchema
- **Model:** 'User'
- The User schema represents user data and includes fields for username, email, password, phone, avatar, role, addresses array, wishlist array, and active status.
- It is used to store user information for registration, authentication, and profile management.
- The email field is marked as unique to ensure each user has a unique email address.
- Passwords are hashed using bcryptjs via a pre-save middleware hook before storing in MongoDB.
- A comparePassword instance method is defined for secure authentication.

```
const userSchema = new mongoose.Schema({
  username: { type: String, required: true, trim: true },
  email: { type: String, required: true, unique: true, lowercase: true },
  password: { type: String, required: true, minlength: 6, select: false },
  phone: { type: String, default: '' },
  role: { type: String, enum: ['user', 'admin'], default: 'user' },
  addresses: [addressSchema],
  wishlist: [{ type: mongoose.Schema.Types.ObjectId, ref: 'Product' }],
  isActive: { type: Boolean, default: true },
}, { timestamps: true });
```

2. Product Schema

- **Schema:** productSchema
- **Model:** 'Product'
- The Product schema represents all pet products available on the platform including food, toys, accessories, and health supplements.
- Key fields include: name, description, price, discountedPrice, discount (%), category, petType (array), brand, images, stock, weight, variants, ratings, tags, isFeatured, isActive, and soldCount.
- A text search index is created on name, description, brand, and tags fields for keyword search functionality.
- A pre-save hook automatically computes the discountedPrice from the base price and discount percentage.

```
const productSchema = new mongoose.Schema({
```

```

name:      { type: String, required: true },
price:     { type: Number, required: true, min: 0 },
discount:  { type: Number, default: 0, min: 0, max: 100 },
category:  { type: String, required: true, enum: [15 categories] },
petType:   { type: [String], enum: ['Dog','Cat','Bird','Fish',...] },
stock:     { type: Number, required: true, min: 0 },
ratings:   { average: Number, count: Number },
isFeatured: { type: Boolean, default: false },
}, { timestamps: true });

```

3. Orders Schema

- **Schema:** orderSchema
- **Model:** 'Order'
- The Order schema stores complete order data including: user reference, order items array (product, name, image, price, quantity, variant), shipping address, payment method and status, pricing breakdown (items, shipping, tax, total), and order status with history.
- The statusHistory field is an array tracking every status change with timestamp and note.
- The orderStatus field follows the lifecycle: Pending → Confirmed → Processing → Shipped → Delivered (or Cancelled/Returned).

```

const orderSchema = new mongoose.Schema({
  user:      { type: ObjectId, ref: 'User', required: true },
  orderItems: [orderItemSchema],
  shippingAddress: shippingAddressSchema,
  paymentMethod: { type: String, enum: ['COD','Card','UPI','NetBanking'] },
  totalPrice:   { type: Number, required: true },
  orderStatus:  { type: String, default: 'Pending' },
  statusHistory: [{ status, timestamp, note }],
}, { timestamps: true });

```

4. Cart Schema

- **Schema:** cartSchema
- **Model:** 'Cart'
- The Cart schema stores the shopping cart data for each user including: user reference (one cart per user), items array (product ref, name, image, price, quantity, variant, stock), and optional coupon/discount fields.
- Each user has exactly one cart document. Items are added/updated/removed in place.

```

const cartSchema = new mongoose.Schema({
  user:      { type: ObjectId, ref: 'User', unique: true },
  items: [{

```

```

    product: { type: ObjectId, ref: 'Product' },
    name:     String,
    price:    Number,
    quantity: { type: Number, min: 1, default: 1 },
    variant:  { type: String, default: '' },
  }],
}, { timestamps: true });

```

5. Admin Schema

- **Schema:** adminSettingsSchema
- **Model:** 'AdminSettings'
- The Admin schema stores site-wide configuration including: banner images for the homepage carousel, product category definitions with icons and subcategories, shipping fee, free shipping threshold, tax rate, store name, tagline, and contact information.

```

const adminSettingsSchema = new mongoose.Schema({
  banners:      [bannerSchema],
  categories:   [categorySchema],
  shippingFee:  { type: Number, default: 50 },
  freeShippingAbove: { type: Number, default: 500 },
  taxRate:      { type: Number, default: 5 },
  siteName:     { type: String, default: 'PetPaws' },
}, { timestamps: true });

```

6. Review Schema

- **Schema:** reviewSchema
- **Model:** 'Review'
- The Review schema stores customer product reviews with: user and product references, rating (1-5), title, comment, helpful votes array, isVerifiedPurchase flag (set automatically based on delivery status), and isApproved status.
- A compound unique index on (user, product) ensures each user can review a product only once.

3. Implement Frontend Logic

Integration with API Endpoints:

- All API calls are made through the centralized `api.js` utility which uses `Axios` with an interceptor to automatically attach the `JWT Authorization` header from `localStorage`.
- A response interceptor handles `401 Unauthorized` errors globally by clearing the token and redirecting to the login page.
- Named API groups (`authAPI`, `productAPI`, `cartAPI`, `orderAPI`, `userAPI`, `wishlistAPI`, `reviewAPI`, `adminAPI`) are exported for clean, module-level usage across all components.

Implement Context Providers:

- **AuthContext:** Provides global user state initialized from `localStorage`. Exposes `login()`, `logout()`, and `updateUser()` functions. On mount, validates the stored token by calling `GET /api/auth/me`.
- **CartContext:** Provides global cart state with `addToCart()`, `updateItem()`, `removeItem()`, `clearCart()` functions. Re-fetches the cart whenever the user state changes. Exposes computed `cartCount` and `cartTotal`.

Route Protection:

- A `ProtectedRoute` component wraps private routes, checking auth state and redirecting unauthenticated users to the login page.
- An `adminOnly` prop on `ProtectedRoute` additionally checks user role and redirects non-admins to the homepage.

PROJECT EXECUTION

Steps for Project Execution

STEP 1: SET UP THE FRONTEND (REACT APP)

- a) Open a terminal and navigate into the client folder:

```
cd client
```

- b) Install all required dependencies:

```
npm install
```

- c) Start the React development server:

```
npm run dev
```

- d) The app should now be running on:

```
http://localhost:5173
```

STEP 2: SET UP THE BACKEND (EXPRESS SERVER)

- a) Open a new terminal tab/window.

- b) Navigate into the server folder:

```
cd server
```

- c) Install all server dependencies:

```
npm install
```

STEP 3: CONFIGURE ENVIRONMENT VARIABLES

- a) Inside the server folder, create a new file named `.env`

- b) Add the following configuration to the `.env` file:

```
PORT=8000
MONGO_URI=mongodb://localhost:27017/petpaws
JWT_SECRET=petpaws_super_secret_jwt_key_2024
JWT_EXPIRE=7d
```



```
NODE_ENV=development
```

STEP 4: SEED THE DATABASE

a) Run the database seeder to populate sample data:

```
npm run seed
```

This will create:

- **Admin Account:** admin@petpaws.com / admin123
- **Test User Account:** john@example.com / user123
- 16 sample pet products across all categories
- Sample delivered order for the test user
- Homepage banner and category configuration

STEP 5: START THE BACKEND SERVER

a) Start the backend with nodemon for hot-reload:

```
npm run dev
```

b) Or for production:

```
npm start
```

c) The backend server starts on:

```
http://localhost:8000
```

d) Verify the API is running:

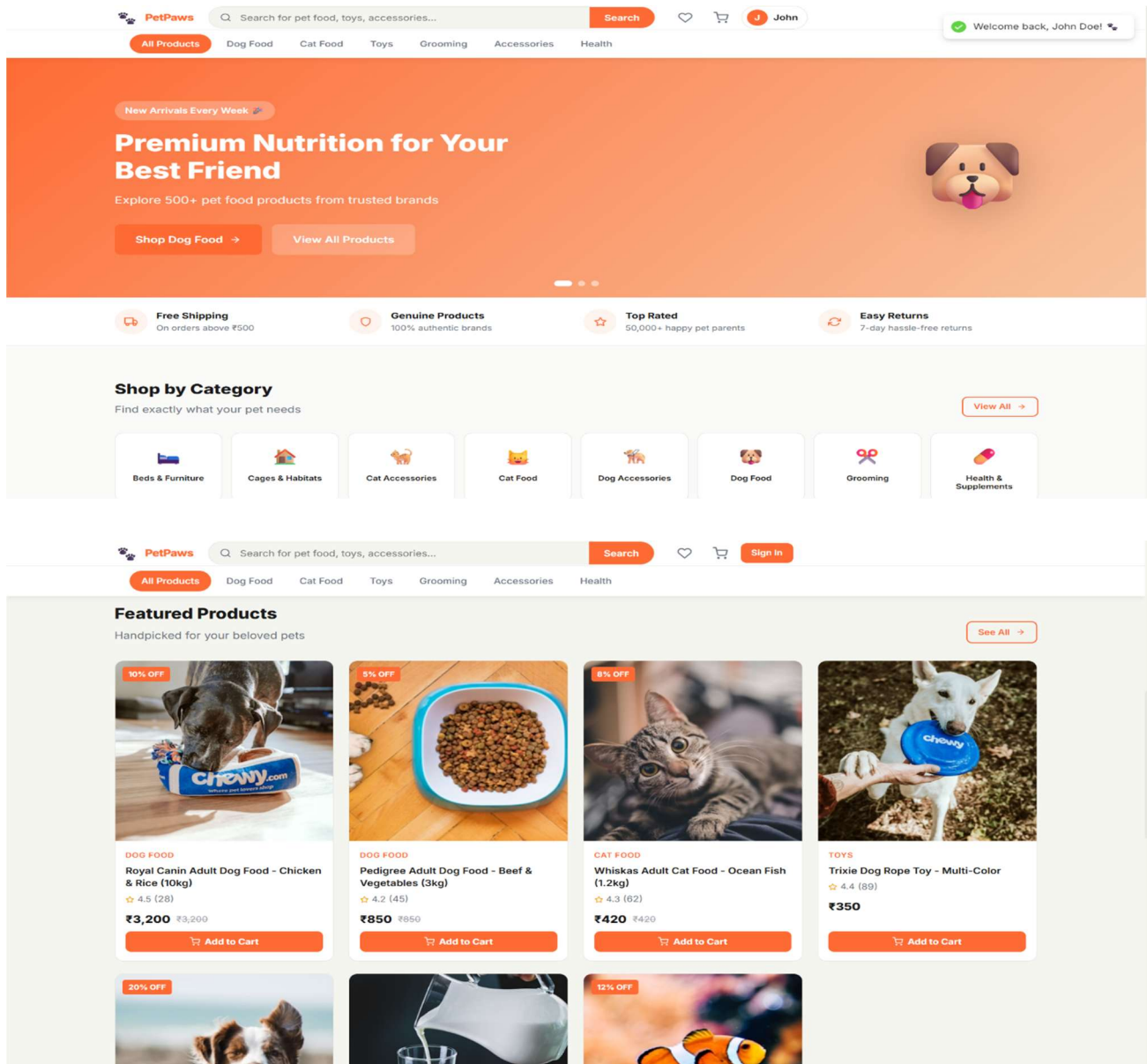
```
http://localhost:8000/api/health
```

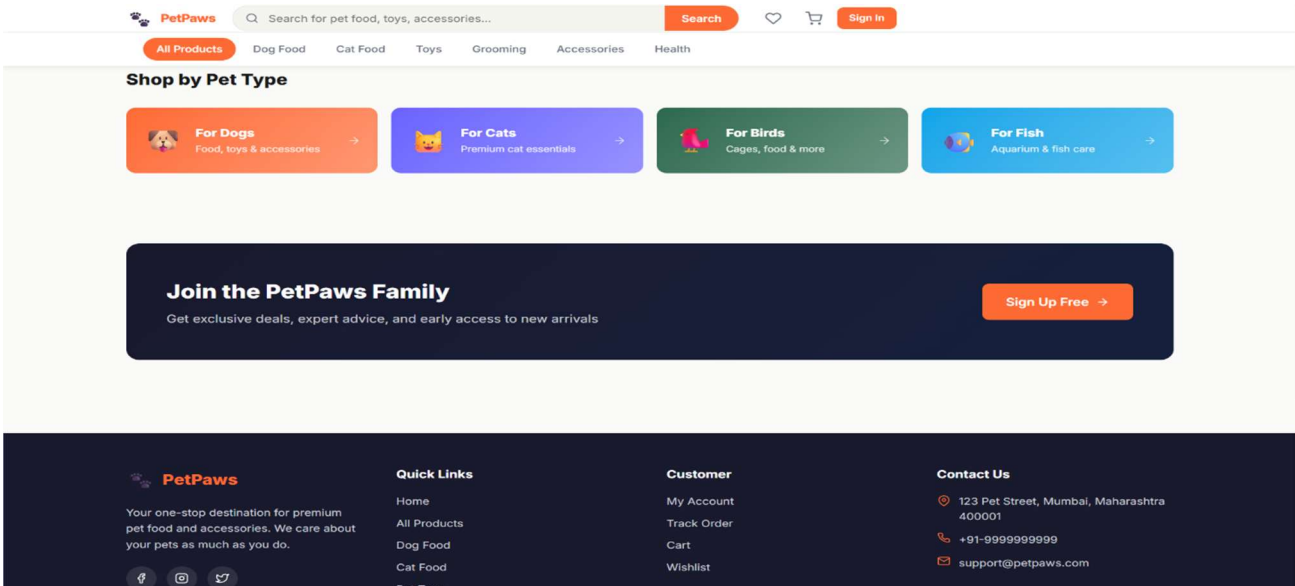
OUTPUT SCREENSHOTS

The following describes each page of the PetPaws application as it appears to the end user and administrator.

Landing Page

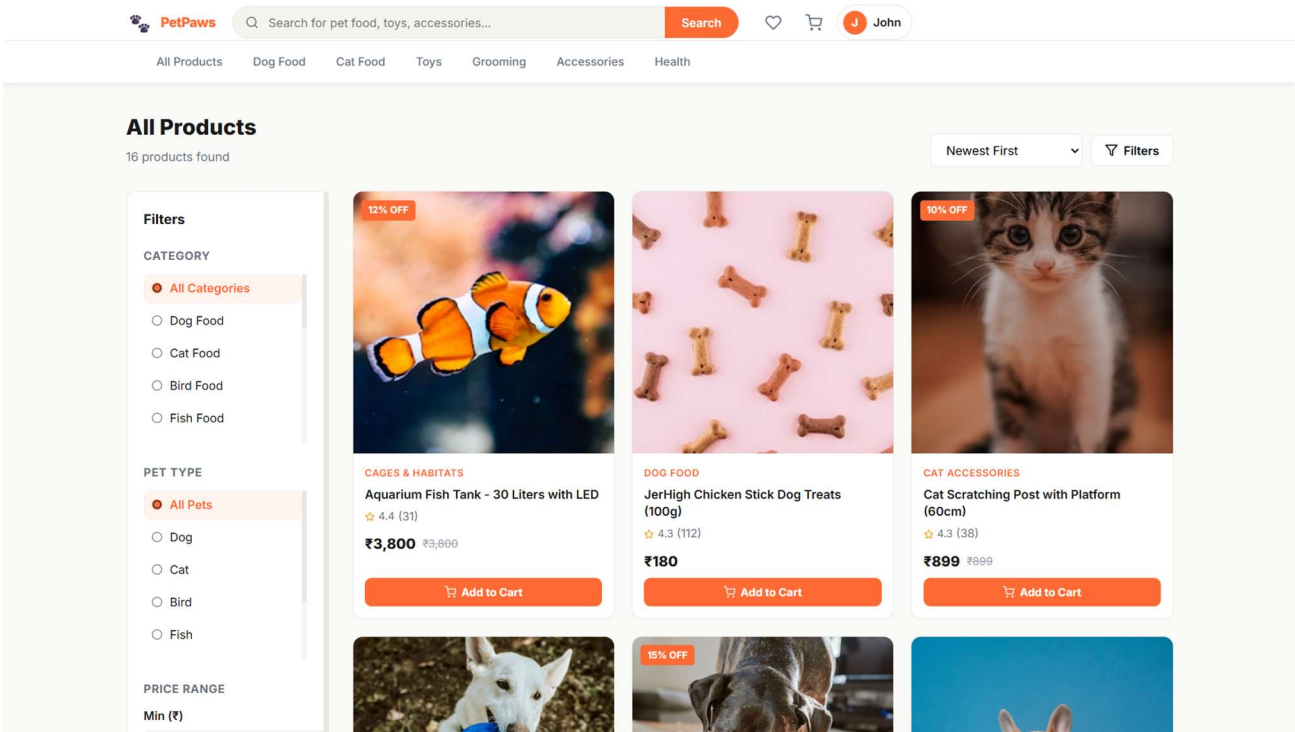
The PetPaws homepage features a full-width animated hero section with a rotating carousel of 3 slides (Pet Food Sale, Grooming Essentials, Cat Products) with floating emoji animations. Below is a trust badges strip (Free Shipping, Genuine Products, Top Rated, Easy Returns), a Shop by Category grid (8 pet categories with emoji icons), a Featured Products grid, a Shop by Pet Type banner row, and a CTA signup section.





Products Page

The Products page displays a responsive product grid with a collapsible sidebar filter panel. Filters include: Category radio list, Pet Type radio list, Price Range with quick-select buttons (Under ₹500, ₹500-₹1000, ₹1000-₹2500, ₹2500+). Products are displayed with a sort dropdown (Newest, Popular, Top Rated, Price Low-High, Price High-Low) and numbered pagination controls at the bottom.



Search
♥
🛒
J John

[All Products](#)
[Dog Food](#)
[Cat Food](#)
[Toys](#)
[Grooming](#)
[Accessories](#)
[Health](#)

Filters

CATEGORY

- ☒ All Categories
- ☐ Dog Food
- ☐ Cat Food
- ☐ Bird Food
- ☐ Fish Food

PET TYPE

- ☒ All Pets
- ☐ Dog
- ☐ Cat
- ☐ Bird
- ☐ Fish

PRICE RANGE

Min (₹)

15% OFF

TOYS

Trixie Dog Rope Toy - Multi-Color

★ 4.4 (89)

₹350

🛒 Add to Cart

15% OFF

BEDS & FURNITURE

Luxury Pet Bed - Plush Memory Foam (Large)

★ 4.6 (15)

₹2,500 ~~₹2,500~~

🛒 Add to Cart

DOG ACCESSORIES

Adjustable Dog Harness - Red (M)

★ 4.5 (67)

₹650

🛒 Add to Cart

12% OFF

5% OFF

Search
♥
🛒
J John

[All Products](#)
[Dog Food](#)
[Cat Food](#)
[Toys](#)
[Grooming](#)
[Accessories](#)
[Health](#)

Cat Food

2 products found

Newest First
🔼 Filters

Filters

✕ Clear All

CATEGORY

- ☐ All Categories
- ☐ Dog Food
- ☒ Cat Food
- ☐ Bird Food
- ☐ Fish Food

PET TYPE

- ☐ All Pets
- ☐ Dog
- ☒ Cat
- ☐ Bird
- ☐ Fish

PRICE RANGE

Min (₹)

8% OFF

CAT FOOD

Whiskas Adult Cat Food - Ocean Fish (1.2kg)

★ 4.3 (62)

₹420 ~~₹420~~

🛒 Add to Cart

12% OFF

CAT FOOD

Purina One Adult Cat Food - Chicken (3kg)

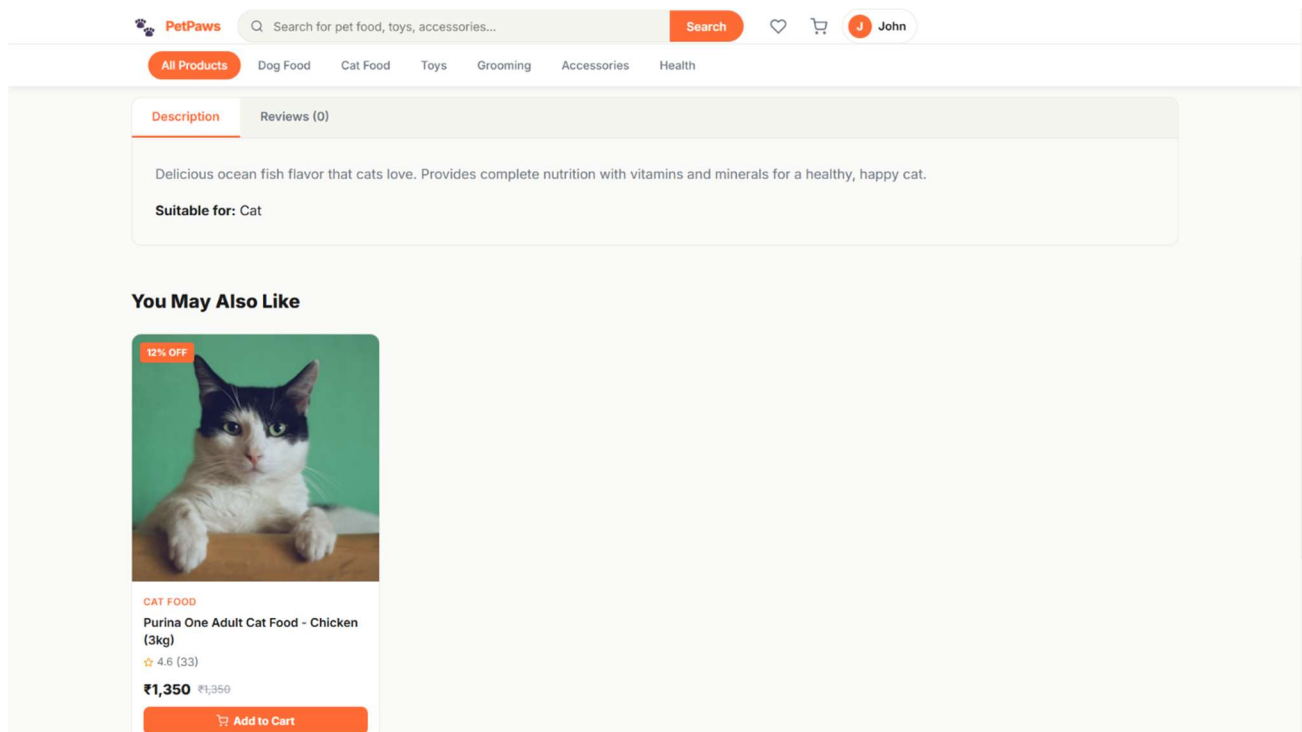
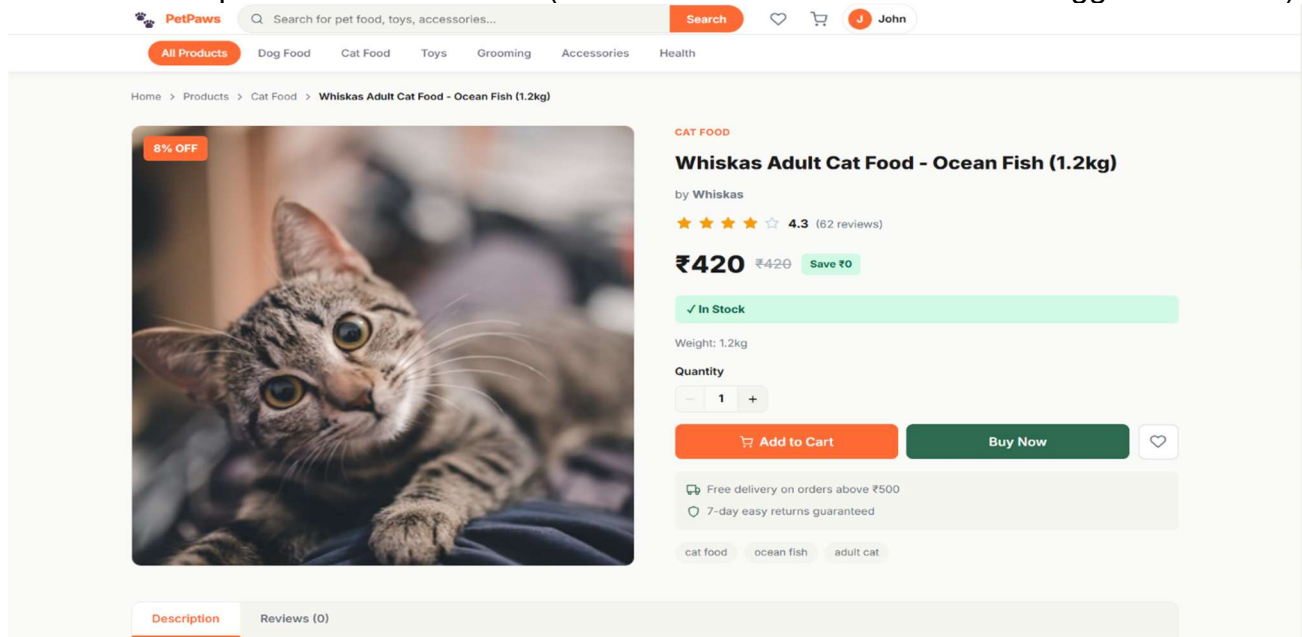
★ 4.6 (33)

₹1,350 ~~₹1,350~~

🛒 Add to Cart

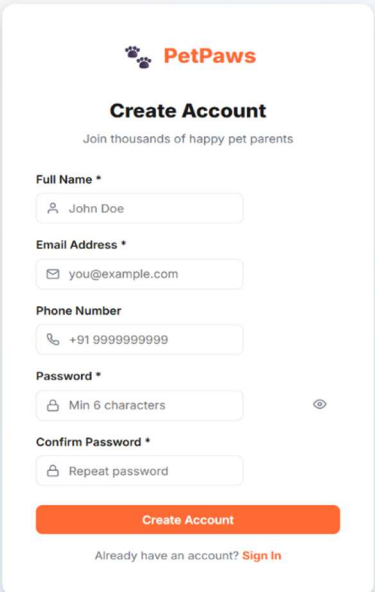
Product Detail Page

The product detail page shows a square main image with thumbnail strip, discount badge, out-of-stock overlay, brand and category info, star rating display, price with savings badge, stock status indicator, variant size/weight selector buttons, quantity stepper, Add to Cart + Buy Now + Wishlist action row, delivery info box, and product tags. A tabbed section below shows Description and Reviews (with a write-review form for logged-in users).

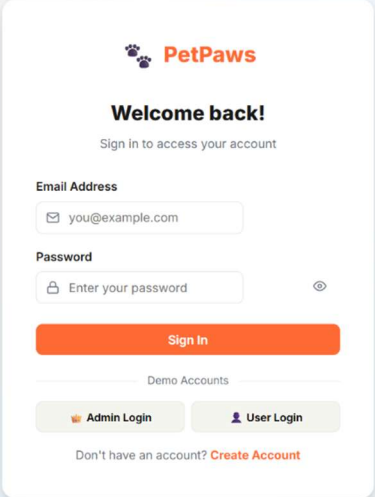


Authentication Pages

The Registration page is a centered card with the PetPaws paw logo, Full Name, Email, Phone, Password (with show/hide toggle), and Confirm Password fields. The Login page adds Demo Account quick-fill buttons for Admin and User accounts for easy testing. Both pages use a gradient background matching the brand colors.



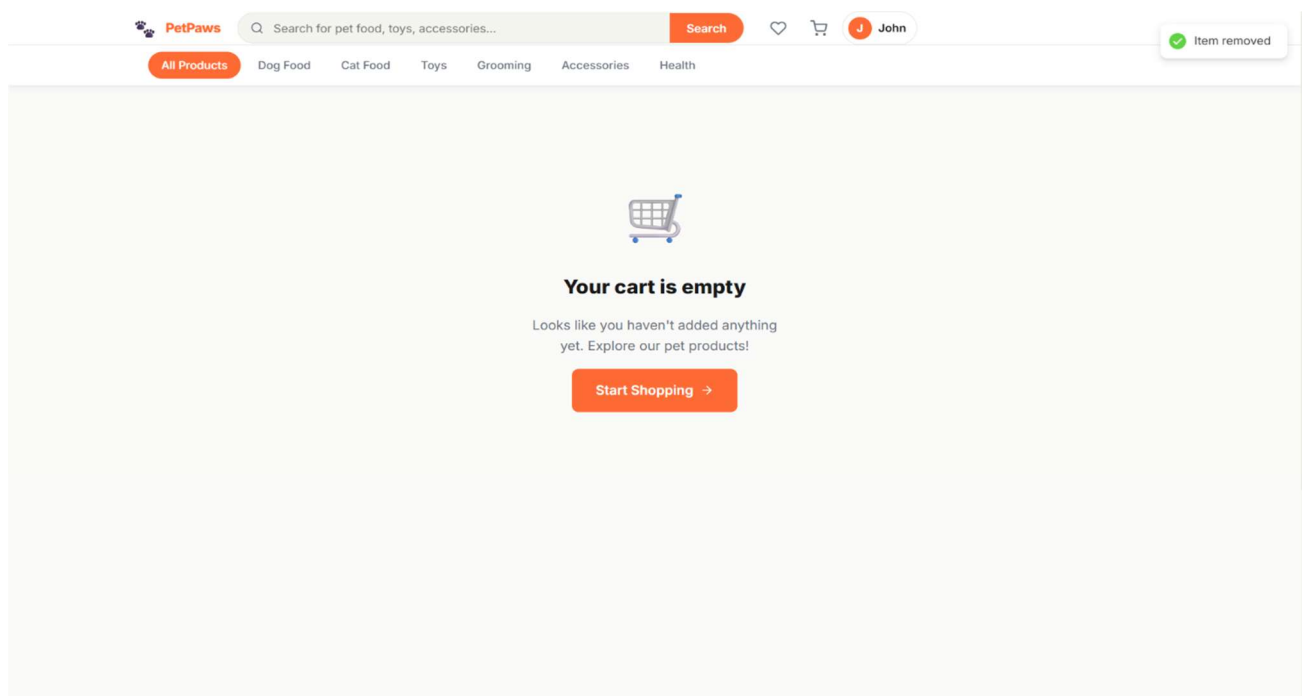
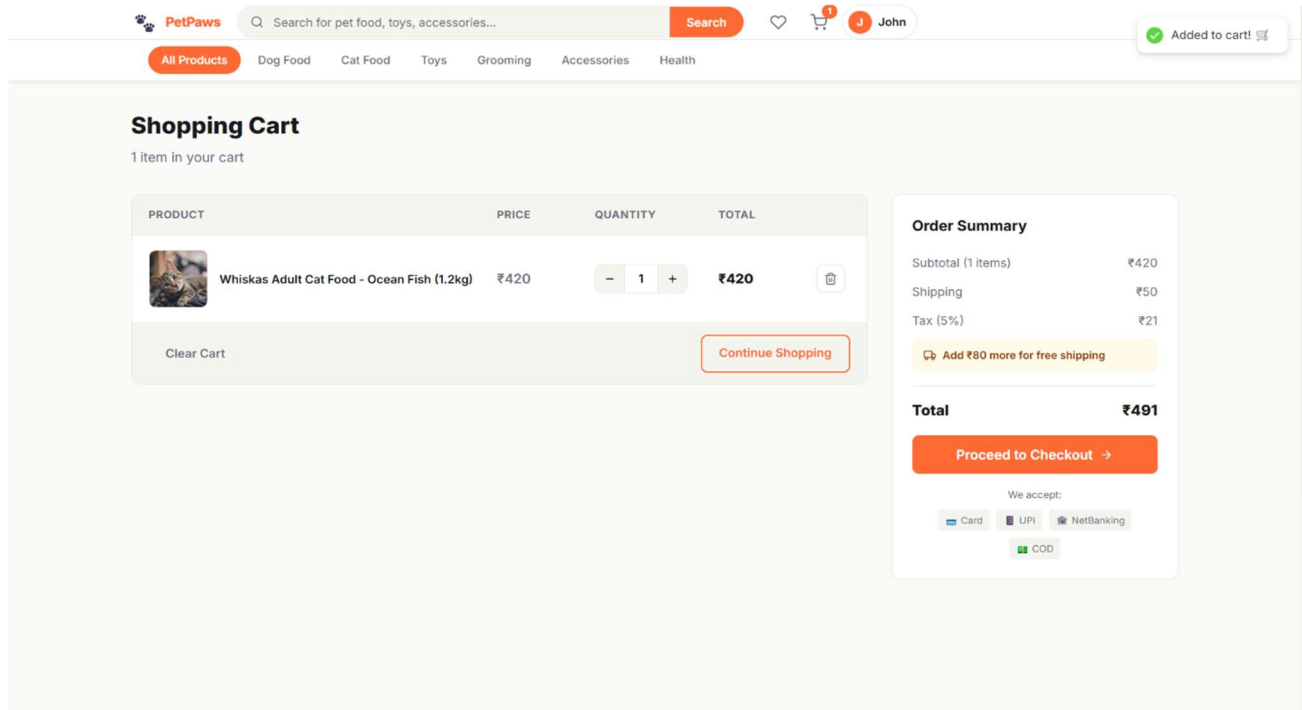
The 'Create Account' form is a white card centered on a light blue and green gradient background. It features the PetPaws logo at the top, followed by the title 'Create Account' and a subtitle 'Join thousands of happy pet parents'. The form includes five input fields: 'Full Name *' (with a person icon), 'Email Address *' (with an email icon), 'Phone Number' (with a phone icon), 'Password *' (with a lock icon and a 'Min 6 characters' hint), and 'Confirm Password *' (with a lock icon and a 'Repeat password' hint). Each field has a small icon to its left. Below the fields is an orange 'Create Account' button. At the bottom, there is a link: 'Already have an account? [Sign In](#)'.



The 'Welcome back! Sign In' form is a white card centered on the same gradient background. It features the PetPaws logo at the top, followed by the title 'Welcome back!' and a subtitle 'Sign in to access your account'. The form includes two input fields: 'Email Address' (with an email icon) and 'Password' (with a lock icon and a hint 'Enter your password'). Below the fields is an orange 'Sign In' button. Underneath the button is a section titled 'Demo Accounts' with two buttons: 'Admin Login' (with a crown icon) and 'User Login' (with a person icon). At the bottom, there is a link: 'Don't have an account? [Create Account](#)'.

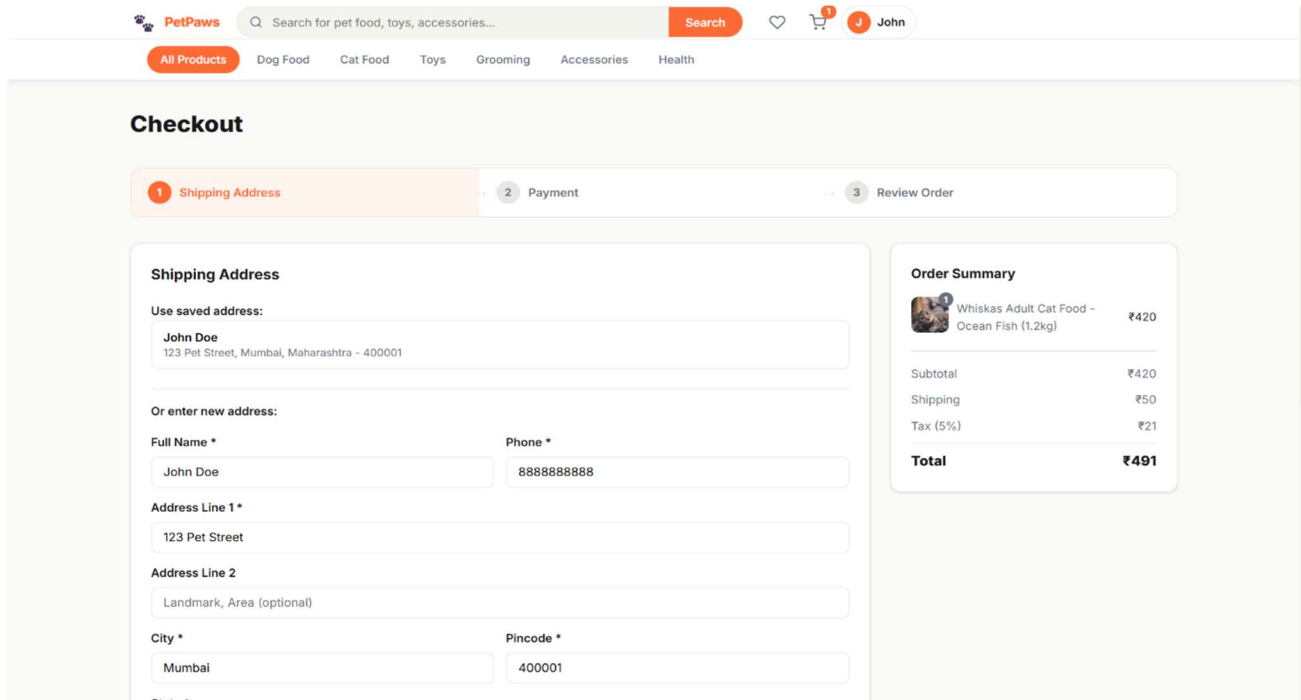
Shopping Cart

The Cart page displays a two-column layout: on the left is a product table with columns for Product (image + name + variant), Price, Quantity stepper, Total, and a Remove button. On the right is the Order Summary card showing subtotal, shipping (FREE if over ₹500), tax (5%), grand total, payment method icons, and the Proceed to Checkout button.



Checkout Page

The Checkout page uses a 3-step progress indicator (Shipping Address → Payment → Review Order). Step 1 shows saved address selector buttons + full address form. Step 2 shows payment method radio cards (COD, Card, UPI, Net Banking) with UPI ID input. Step 3 shows the complete order review with edit links. The right sidebar shows a persistent order summary throughout all steps.



Checkout

1 Shipping Address → 2 Payment → 3 Review Order

Shipping Address

Use saved address:

John Doe
123 Pet Street, Mumbai, Maharashtra - 400001

Or enter new address:

Full Name * Phone *

John Doe 8888888888

Address Line 1 *
123 Pet Street

Address Line 2
Landmark, Area (optional)

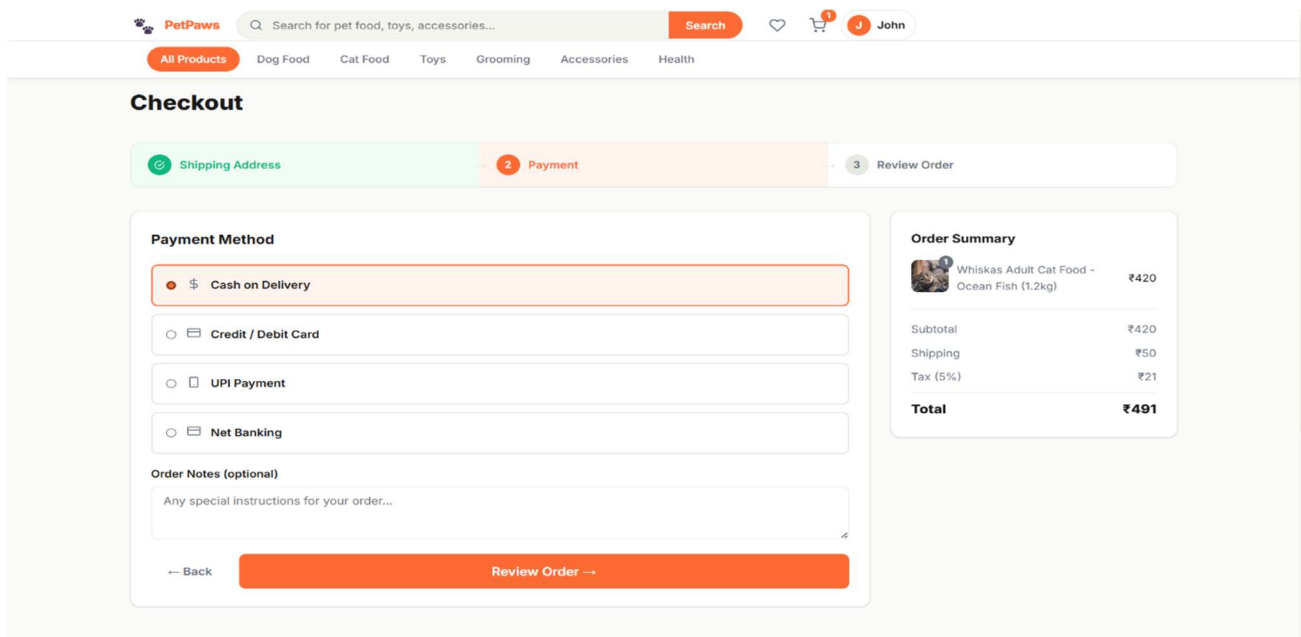
City * Pincode *

Mumbai 400001

Order Summary

Whiskas Adult Cat Food - Ocean Fish (1.2kg) ₹420

Subtotal ₹420
Shipping ₹50
Tax (5%) ₹21
Total ₹491



Checkout

1 Shipping Address → 2 Payment → 3 Review Order

Payment Method

☒ Cash on Delivery

☐ Credit / Debit Card

☐ UPI Payment

☐ Net Banking

Order Notes (optional)

Any special instructions for your order...

← Back Review Order →

Order Summary

Whiskas Adult Cat Food - Ocean Fish (1.2kg) ₹420

Subtotal ₹420
Shipping ₹50
Tax (5%) ₹21
Total ₹491

Search
♥
🛒 1
J John

All Products
Dog Food
Cat Food
Toys
Grooming
Accessories
Health

Checkout

🏠 Shipping Address
→ 🏠 Payment
→ 3 Review Order

Review Your Order

SHIPPING TO Edit

John Doe - 8888888888
 123 Pet Street
 Mumbai, Maharashtra - 400001

PAYMENT Edit

Cash on Delivery

ITEMS (1)

Whiskas Adult Cat Food - Ocean Fish (1.2kg)

×1 ₹420

← Back
Place Order - ₹491

Order Summary

Whiskas Adult Cat Food - Ocean Fish (1.2kg)

₹420

Subtotal

₹420

Shipping

₹50

Tax (5%)

₹21

Total

₹491

Search
♥
🛒 1
J John

All Products
Dog Food
Cat Food
Toys
Grooming
Accessories
Health

🟢 Order placed successfully! 🎉

Order Confirmed! 🎉

Thank you for your order. Your order ID is #F01DD3D9

Order Details

Whiskas Adult Cat Food - Ocean Fish (1.2kg)
 Qty: 1

₹420

Total Paid

₹491

Payment: COD - Status: Pending

📍 Track Order
📄 All Orders
🛒 Continue Shopping

User Profile

The Profile page has a 3-tab layout: (1) Personal Information — editable name and phone fields, read-only email. (2) Password — current password + new password + confirm password with validation. (3) Addresses — saved address cards with Edit/Delete buttons and an Add Address form with full Indian state dropdown.

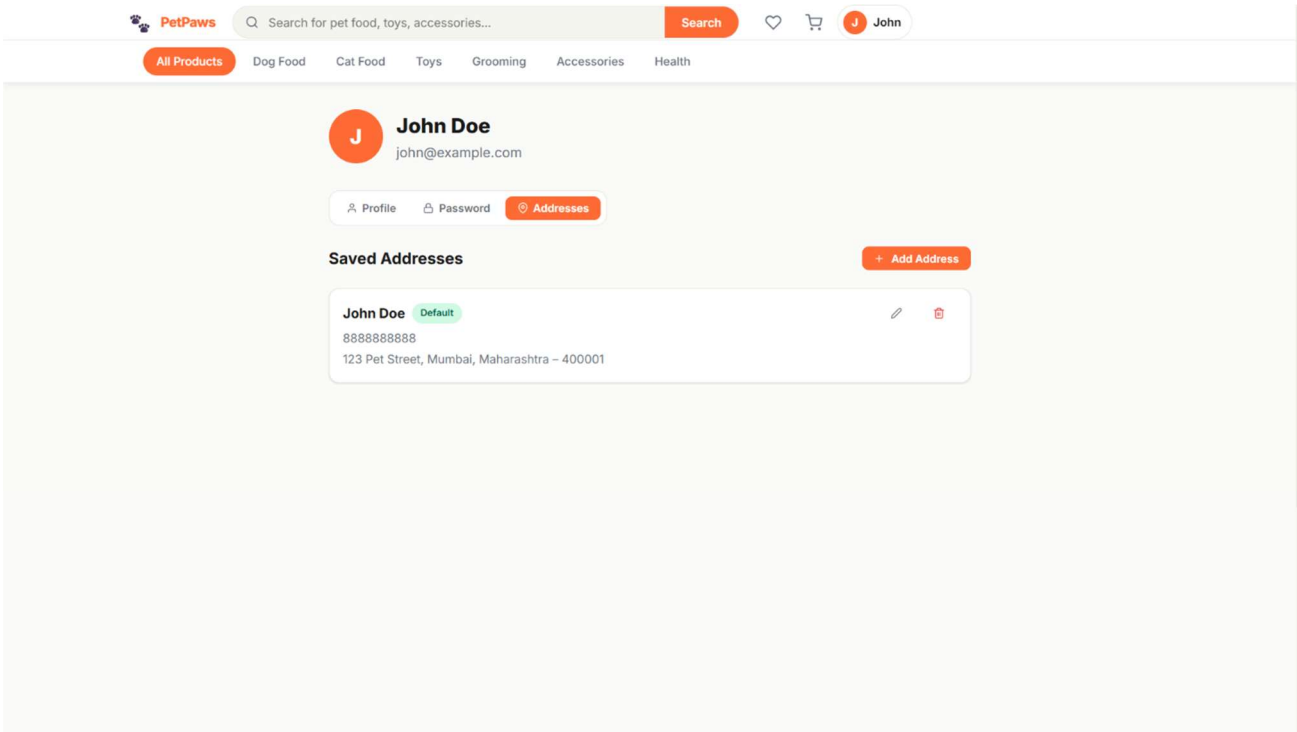
The image displays two screenshots of the PetPaws user profile page, illustrating the 3-tab layout described in the text.

Top Screenshot: Personal Information Tab

- Header:** PetPaws logo, search bar (Search for pet food, toys, accessories...), Search button, user profile (John Doe), and navigation links (All Products, Dog Food, Cat Food, Toys, Grooming, Accessories, Health).
- User Profile:** John Doe, john@example.com.
- Tabs:** Profile (active), Password, Addresses.
- Form Fields:**
 - Full Name: John Doe
 - Email (cannot change): john@example.com
 - Phone: 8888888888
- Action:** Save Changes button.

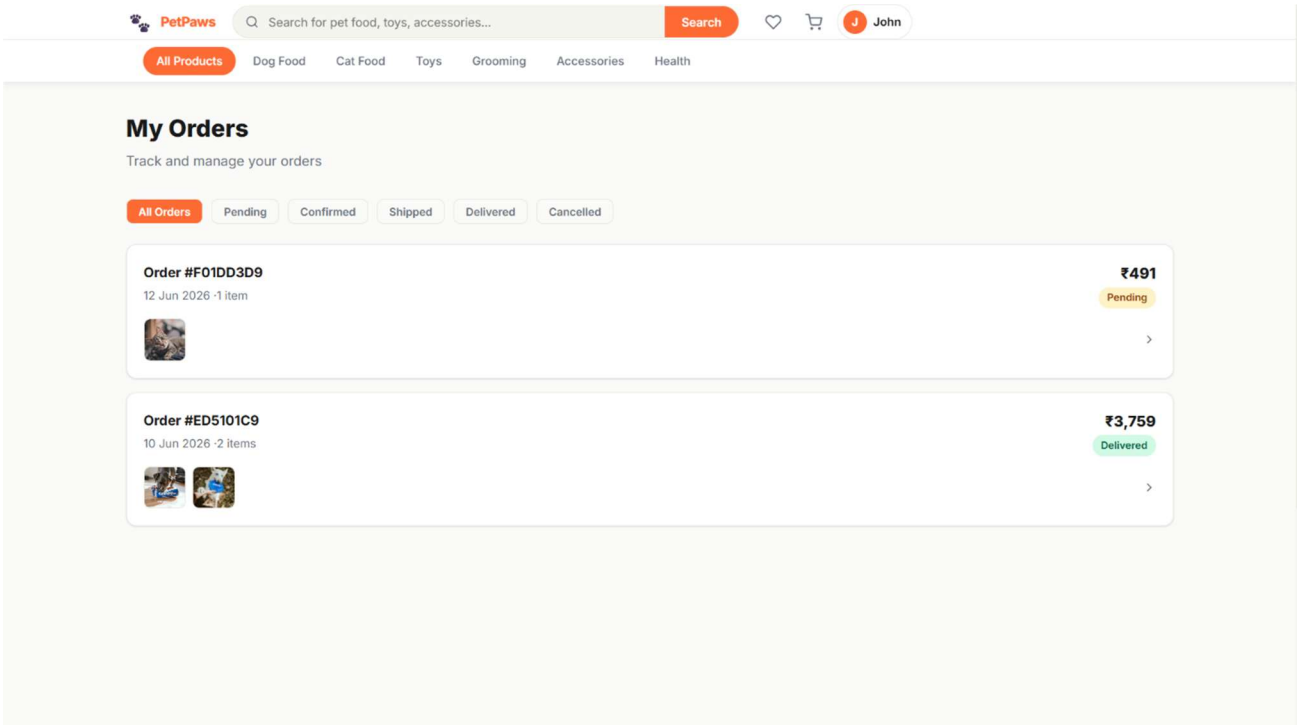
Bottom Screenshot: Password Tab

- Header:** Same as the top screenshot.
- User Profile:** John Doe, john@example.com.
- Tabs:** Profile, Password (active), Addresses.
- Form Fields:**
 - Current Password
 - New Password
 - Confirm New Password
- Action:** Update Password button.



My Orders

The Orders page shows a list of order cards with status filter tabs (All, Pending, Confirmed, Shipped, Delivered, Cancelled). Each card shows Order ID, date, item count, total, product image thumbnails (up to 4), and a colour-coded status badge. Clicking any card navigates to the Order Detail page.



Search
♥
🛒
J John

All Products
Dog Food
Cat Food
Toys
Grooming
Accessories
Health

Order #F01DD3D9

Placed on 12 June 2026

Pending
Cancel Order

Order Status

Items Ordered

Whiskas Adult Cat Food - Ocean Fish (1.2kg)
₹420 × 1

₹420

Shipping Address

John Doe
8888888888
123 Pet Street
Mumbai, Maharashtra - 400001

Payment Details

Method: COD
Status: Pending

Subtotal	₹420
Shipping	₹50
Tax	₹21
Total	₹491

Search
♥
🛒
J John

All Products
Dog Food
Cat Food
Toys
Grooming
Accessories
Health

Order #ED5101C9

Placed on 10 June 2026

Delivered

Order Status

Items Ordered

Royal Canin Adult Dog Food - Chicken & Rice (10kg)
₹2,880 × 1

₹2,880

Trixie Dog Rope Toy - Multi-Color
₹350 × 2

₹700

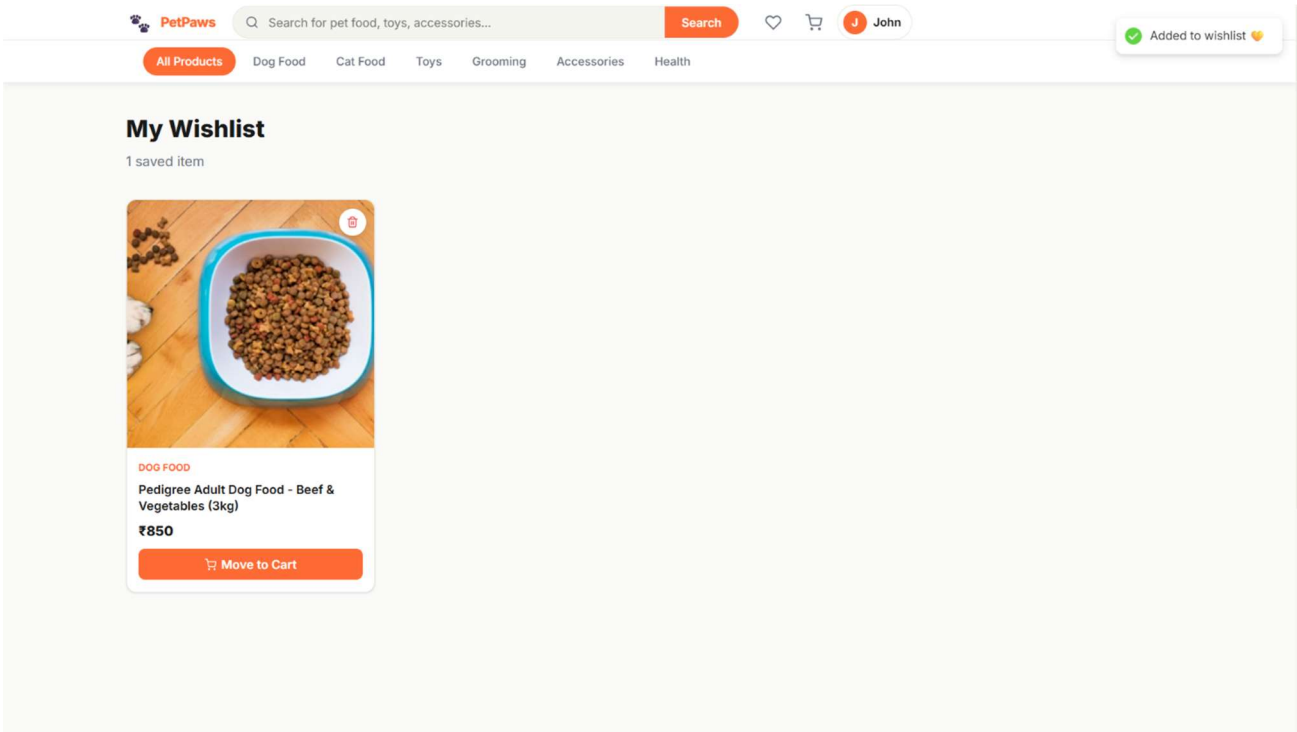
Shipping Address

John Doe
8888888888
123 Pet Street
Mumbai, Maharashtra - 400001

Payment Details

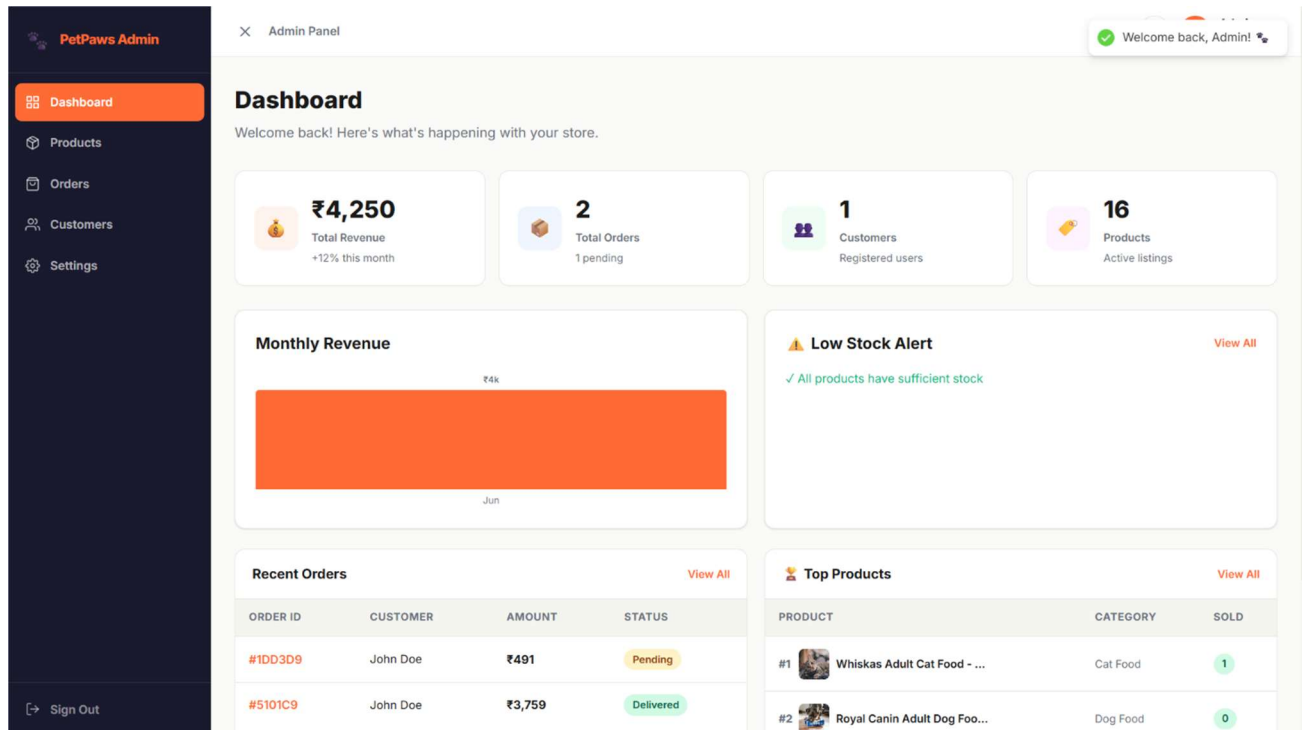
Method: COD
Status: Pending

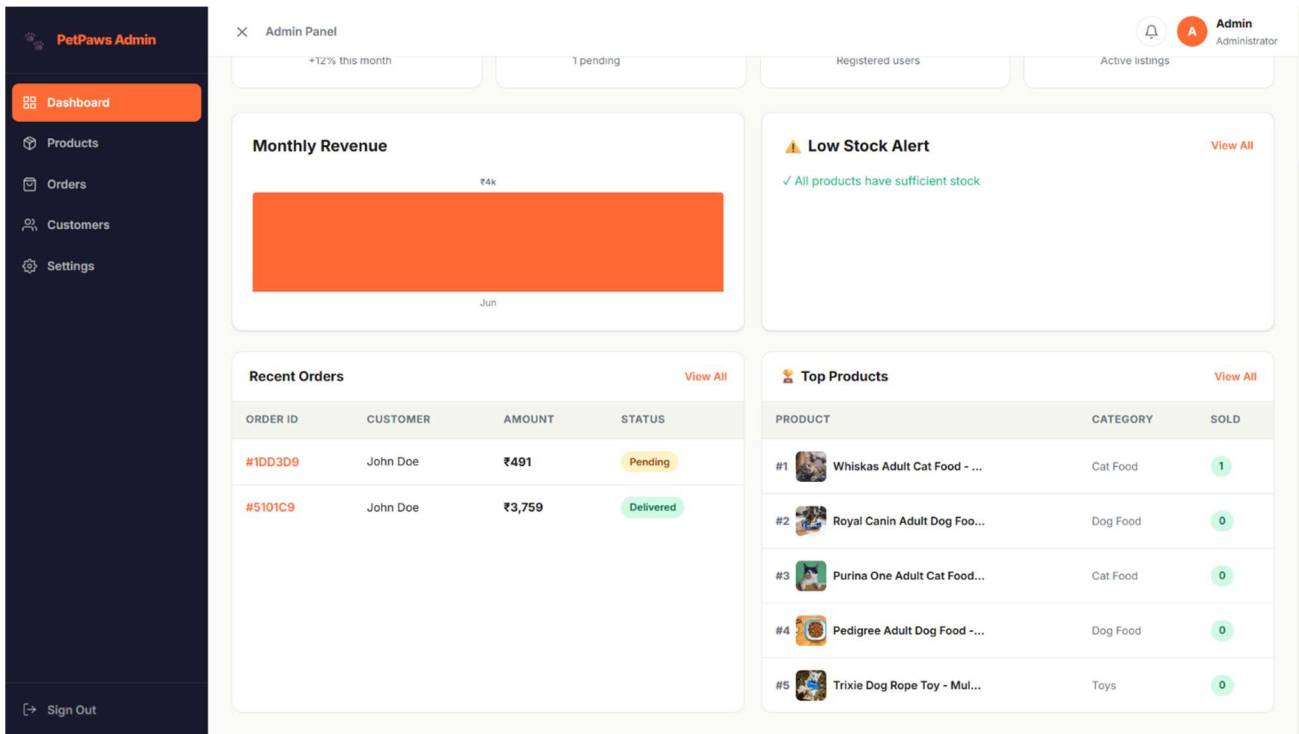
Subtotal	₹3,580
Shipping	FREE
Tax	₹179



Admin Dashboard

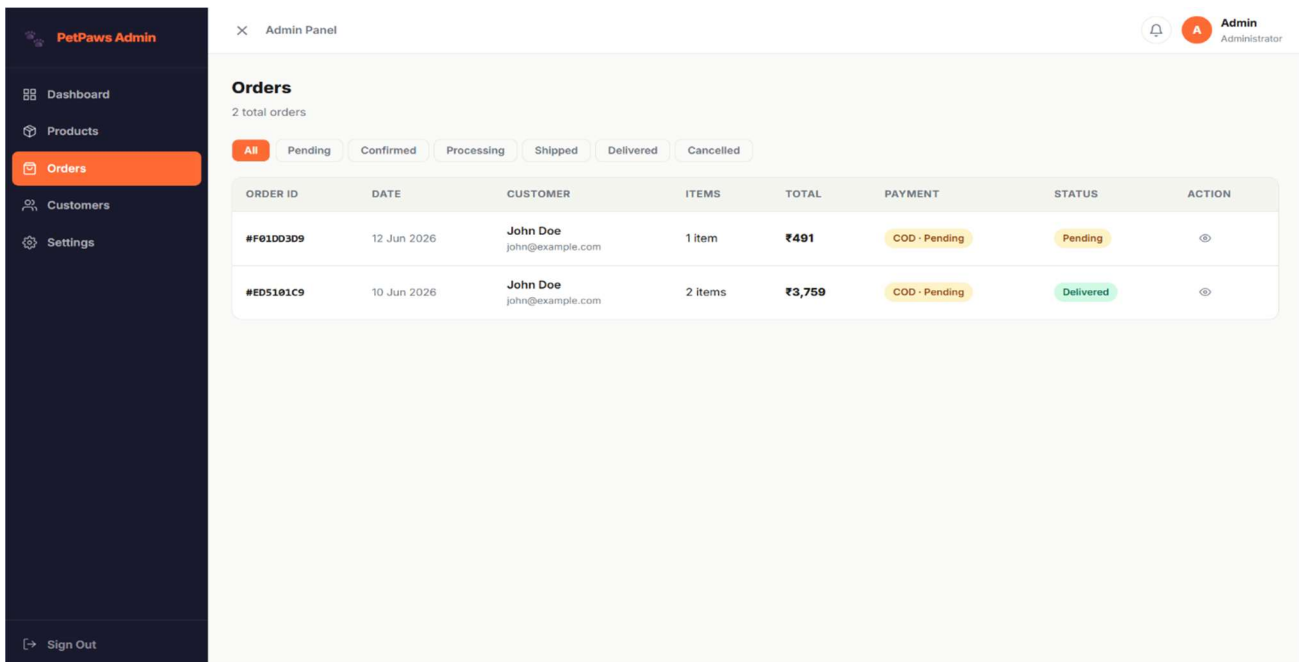
The Admin Dashboard shows 4 KPI stat cards (Total Revenue, Total Orders, Customers, Products), a monthly revenue bar chart (bar heights proportional to revenue per month), a Low Stock Alert table, a Recent Orders quick table, and a Top Products ranking table with sold count badges.





Admin Orders

The All Orders admin page shows a comprehensive table with columns: Order ID, Date, Customer (name + email), Items count, Total, Payment (method + status), Order Status badge, and a View Details eye button. Status filter tabs at the top allow filtering by order state. Each order links to a detailed management page with status update form, tracking number input, and timeline history.



PetPaws Admin

Dashboard
Products
Orders
Customers
Settings

Sign Out

Admin Panel

Order #F01DD3D9

Placed on 12 June 2026 at 05:52 pm

Pending

Order Items

Whiskas Adult Cat Food - Ocean Fish (1.2kg)

₹420 × 1

₹420

Customer

John Doe
john@example.com
8888888888

Shipping Address

John Doe
8888888888
123 Pet Street, Mumbai, Maharashtra - 400001

Status History

Pending

Order placed

12/6/2026, 5:52:04 pm

Update Order Status

Status

Pending

Tracking Number

e.g., BD123456789

Note

Optional note...

Update Status

Payment Summary

Subtotal

₹420

Shipping

₹50

Tax

₹21

Total

₹491

COD - Pending

Admin Products

The All Products admin page shows a searchable product table with columns: Product (image + name + brand), Category badge, Price (with crossed-out original if discounted), Stock badge (green/yellow/red), Rating, Featured status badge, and Edit/View/Delete action buttons. An Add Product button navigates to the product form.

PetPaws Admin

Dashboard
Products
Orders
Customers
Settings

Sign Out

Admin Panel

Products

16 total products

Search products...

Search

All Categories

PRODUCT	CATEGORY	PRICE	STOCK	RATING	FEATURED	ACTIONS
AquaWorld	Cages & Habitats	₹3,800 ₹3,800	20 units	★ 4.4 (31)	✓ Featured	View Edit Delete
JerHigh	Dog Food	₹180 ₹180	200 units	★ 4.3 (112)	Not Featured	View Edit Delete
CatComfort	Cat Accessories	₹899 ₹699	35 units	★ 4.3 (38)	Not Featured	View Edit Delete
Trixie	Toys	₹350 ₹350	120 units	★ 4.4 (89)	✓ Featured	View Edit Delete
DreamPet	Beds & Furniture	₹2,500 ₹2,500	25 units	★ 4.6 (15)	Not Featured	View Edit Delete
PawsUp	Dog Accessories	₹650 ₹650	75 units	★ 4.5 (67)	Not Featured	View Edit Delete
Purina	Cat Food	₹1,350 ₹1,350	65 units	★ 4.6 (33)	Not Featured	View Edit Delete
VetCare	Health & Supplements	₹750 ₹750	60 units	★ 4.8 (44)	✓ Featured	View Edit Delete

New Product Page

The Add/Edit Product form is a two-column layout. The left column contains: Basic Information (name, description, brand, weight, tags), Pricing & Stock (price, discount %, stock quantity with live discounted price preview), and Images (main image URL + additional images gallery). The right column contains: Category dropdown, Pet Type multi-select toggle buttons, and Settings (Featured toggle, Active/Visible toggle).

PetPaws Admin

Admin Panel

Dashboard

Products

Orders

Customers

Settings

Sign Out

Back **Add New Product** Save Product

Basic Information

Product Name *
e.g., Royal Canin Adult Dog Food 10kg

Description *
Describe the product...

Brand
e.g., Royal Canin

Weight / Size
e.g., 10kg, 500ml

Tags (comma separated)
e.g., dog food, premium, chicken

Pricing & Stock

Price (₹) *
0

Discount (%)
0

Stock Qty *
0

Category

Product Category *
Dog Food

Suitable For

Dog Cat Bird Fish
Rabbit Hamster Reptile
All

Settings

☐ Featured Product
Show on homepage

☒ Active / Visible
Product is listed publicly

Create Product

Images

Main Image URL *
https://...

Additional Images
Image URL... +

Admin Customers

The **Admin Customers** page displays all registered customers in a searchable table. It includes a customer search bar for filtering by name or email. The customer table shows Customer Details (name, email, avatar), Phone Number, Registration Date, Last Login Date, Number of Saved Addresses, Account Status (Active/Inactive), and Action buttons. Administrators can view customer information, monitor activity, and activate/deactivate customer accounts directly from the table.

PetPaws Admin

Admin Panel

Customers
1 registered customers

Search by name or email... **Search**

CUSTOMER	PHONE	JOINED	LAST LOGIN	ADDRESSES	STATUS	ACTIONS
 John Doe john@example.com	8888888888	10 Jun 2026	12/6/2026	1	Active	Deactivate

Sign Out

Admin Settings

The Admin Settings page provides a centralized interface for managing store-wide configurations and homepage content. Administrators can update general store information, including the store name, tagline, contact email, and phone number. The system allows configuration of shipping fees, free shipping thresholds, and applicable tax rates. Shipping and tax policies can be reviewed and modified directly from the settings panel. Homepage banners can be added, edited, enabled, disabled, or removed to manage promotional content. Banner images and redirect links can also be configured to enhance customer engagement. The category management section allows administrators to view and control product category availability. Categories can be enabled or disabled to determine their visibility on the storefront. Summary cards provide a quick overview of store information, shipping settings, active banners, and category status. All configuration changes can be saved through the settings panel, ensuring efficient administration and consistent store management.

PetPaws Admin

Dashboard
Products
Orders
Customers
Settings

Sign Out

Admin Panel

Store Settings

Save Settings

General Info

Store Name
PetPaws

Tagline
Everything your pet needs

Contact Email
support@petpaws.com

Contact Phone
+91-9999999999

Shipping & Tax

Shipping Fee (₹)
50

Free Shipping Above (₹)
500

Tax Rate (%)
5

Orders above ₹500 get free shipping. Others pay ₹50. Tax of 5% is applied on all orders.

Homepage Banners

Add Banner

Premium Pet Food Sale

Up to 30% off on Royal Canin & Whiskas

/products?category=Dog+Food

Active

Grooming Essentials

Everything your pet needs to look & feel great

/products?category=Grooming

Active

PetPaws Admin

Dashboard
Products
Orders
Customers
Settings

Sign Out

Admin Panel

Tagline

Everything your pet needs

Contact Email

support@petpaws.com

Contact Phone

+91-9999999999

Free Shipping Above (₹)

500

Tax Rate (%)

5

Orders above ₹500 get free shipping. Others pay ₹50. Tax of 5% is applied on all orders.

Homepage Banners

Add Banner

Premium Pet Food Sale

Up to 30% off on Royal Canin & Whiskas

/products?category=Dog+Food

Active

Grooming Essentials

Everything your pet needs to look & feel great

/products?category=Grooming

Active

Configured Categories (8)

Dog Food On

Cat Food On

Toys On

Grooming On

Dog Accessories On

Cat Accessories On

Health & Supplements On

Beds & Furniture On

API DOCUMENTATION

Authentication Endpoints

Method Endpoint	Description
POST /api/auth/register	Register a new user. Body: { username, email, password, phone }
POST /api/auth/login	Login user. Body: { email, password }. Returns JWT token.
POST /api/auth/admin/login	Admin-specific login. Body: { email, password }.
GET /api/auth/me	Get current authenticated user profile. Requires JWT.

Product Endpoints

Method Endpoint	Description
GET /api/products	Get all products. Query: keyword, category, petType, brand, minPrice, maxPrice, rating, sort, page, limit, featured.
GET /api/products/featured	Get featured products for homepage display.
GET /api/products/brands	Get a distinct list of all product brand names.
GET /api/products/:id	Get single product details + reviews by product ID.
GET /api/products/:id/related	Get related products from the same category.
POST /api/products	Create a new product. Admin only.
PUT /api/products/:id	Update an existing product. Admin only.
DELETE /api/products/:id	Soft-delete a product (sets isActive: false). Admin only.

Cart Endpoints

Method Endpoint	Description
GET /api/cart	Get the authenticated user's cart with populated product details.
POST /api/cart	Add a product to the cart. Body: { productId, quantity, variant }.
PUT /api/cart/:itemId	Update quantity of a specific cart item. Body: { quantity }.
DELETE /api/cart/:itemId	Remove a specific item from the cart by item ID.
DELETE /api/cart	Clear all items from the user's cart.

Order Endpoints

Method Endpoint	Description
POST /api/orders	Place a new order. Validates stock, computes totals, deducts stock.
GET /api/orders/my	Get the authenticated user's order history. Query: status, page, limit.
GET /api/orders/:id	Get detailed information for a specific order.
PUT /api/orders/:id/cancel	Cancel an eligible order (Pending or Confirmed only).
GET /api/orders	Get all orders. Admin only. Query: status, page, limit.
PUT /api/orders/:id/status	Update order status. Admin only. Body: { status, note, trackingNumber }.

CONCLUSION

PetPaws successfully demonstrates a complete, production-ready MERN stack e-commerce application tailored for the pet foods and accessories domain. The project covers all the essential modules of a modern online store — from product discovery and cart management to secure checkout, order tracking, and a comprehensive admin panel.

The application is built following industry best practices including the MVC architectural pattern, JWT-based stateless authentication, MongoDB aggregation for analytics, React Context API for global state management, and responsive CSS design with mobile-first considerations.

Key achievements of the PetPaws project:

- **Full-Stack Integration:** Seamless communication between the React frontend and the Express.js backend via Axios with JWT interceptors.
- **Domain Specificity:** All 15 product categories, 8 pet types, and sample data are specifically designed for the pet store domain.
- **Security:** Passwords are bcrypt-hashed, routes are protected with JWT middleware, and admin routes have role-based access control.
- **Scalability:** The modular folder structure (models, controllers, routes) makes it straightforward to add new features, categories, or payment providers.
- **User Experience:** From animated hero carousels to hover-effect product cards, multi-step checkout, and real-time order tracking — the UI prioritizes a smooth and delightful shopping experience for pet owners.

PetPaws can be further extended with features such as real payment gateway integration (Razorpay, PayU), email notifications via Nodemailer, product image upload via Cloudinary, and deployment on platforms like Vercel (frontend) and Railway (backend) for a fully live production application.

Demo Credentials:

Role	Email Password
Administrator	admin@petpaws.com admin123
Registered User	john@example.com user123

Project Code & Source Files:

- Full source code included in the project ZIP archive: petpaws-mern-app.zip
- Backend: /server — Node.js + Express.js + MongoDB
- Frontend: /client — React.js + Vite
- Seed script: /server/data/seeder.js — Run with: npm run seed